BECE102L – Digital Systems Design

Module 5 –Design of Sequential Logic Circuits

Latches, Flip-Flops - SR, D, JK & T, Buffer Registers, Shift Registers - SISO, SIPO, PISO, PIPO, Design of synchronous sequential circuits: state table and state diagrams, Design of counters: Modulo-n, Johnson, Ring, Up/Down, Asynchronous counter. Modeling of sequential logic circuits using Verilog HDL..

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flip-flop conversions

Follow these steps for converting one flip-flop to the other.

1. Consider the characteristic table of desired flip-flop.
2. Fill the excitation values inputs of given flip-flop for each combination of present state and next state.
3. Get the simplified expressions for each excitation input. If necessary, use K maps for simplifying.
4. Draw the circuit diagram of desired flip-flop according to the simplified expressions using given flip-flop and necessary logic gates.

Excitation Table for all flip-flops

Present State	Next State	SR flip-flop inputs		D flip-flop input	JK flip-flop inputs		T flip-flop input
Q t	Q $t+1$	S	R	D	J	K	T
0	0	0	x	0	0	x	0
0	1	1	0	1	1	x	1
1	0	0	1	0	x	1	1
1	1	x	0	1	x	0	0

SR flip-flop to D flip-flop conversion

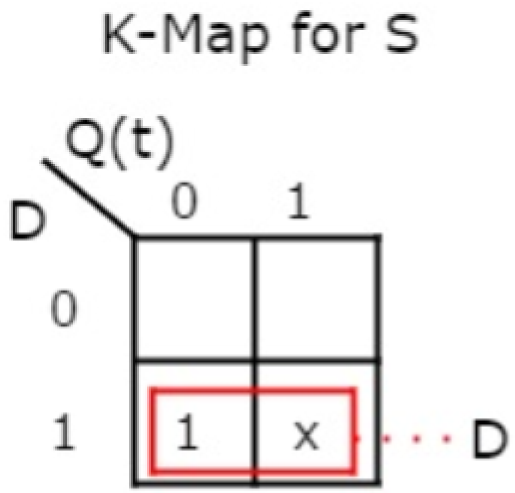
Characteristic table of D flip-flop

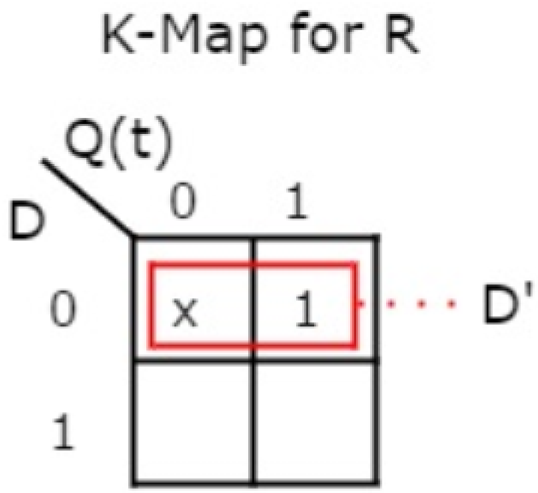
D flip-flop input	Present State	Next State
D	Q t	Q $t+1$
0	0	0
0	1	0
1	0	1
1	1	1

Characteristic table of D flip-flop along with the **excitation inputs of SR flip-flop.**

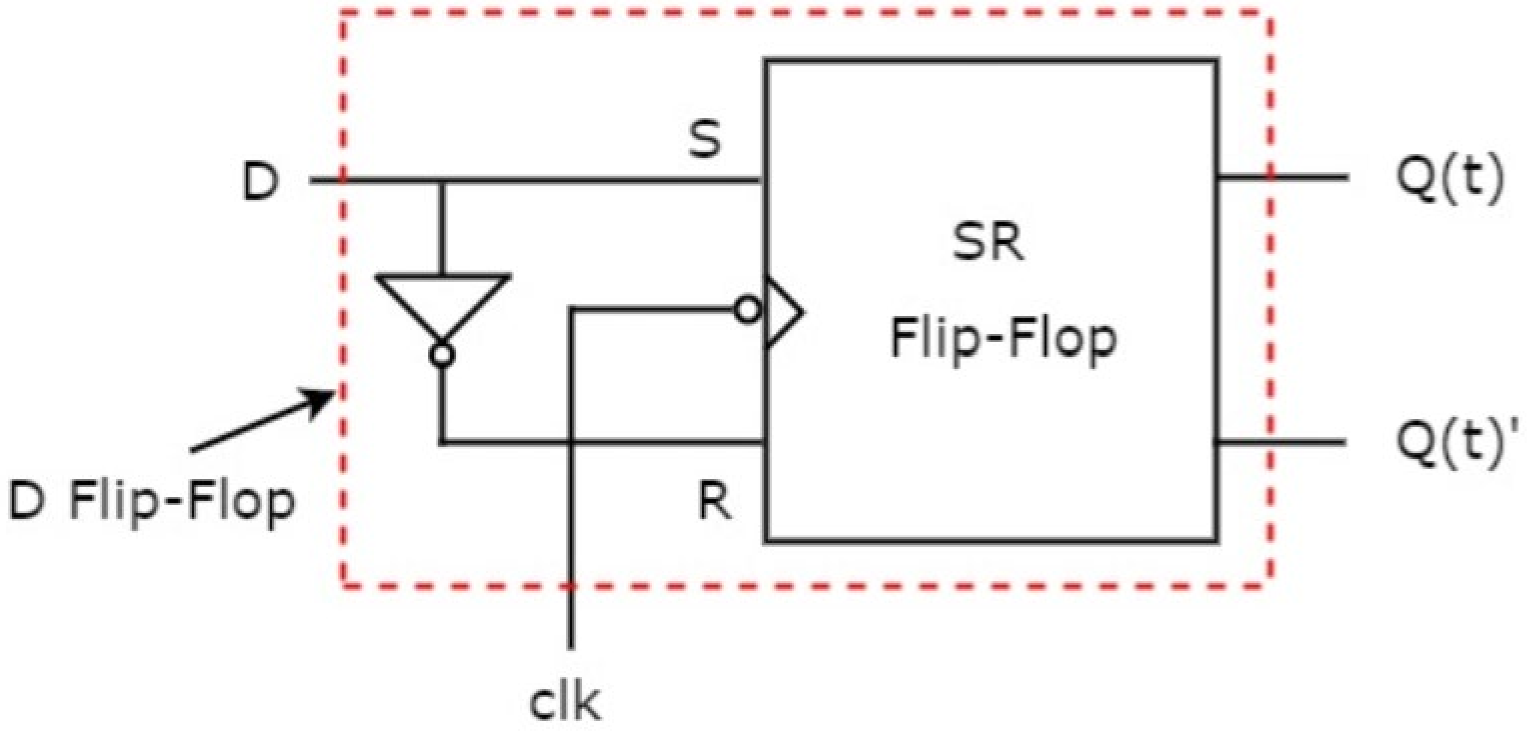
D flip-flop input	Present State	Next State	SR flip-flop inputs	
D	Q t	Q $t+1$	S	R
0	0	0	0	x
0	1	0	0	1
1	0	1	1	0
1	1	1	x	0

We can use 2 variable K-Maps for getting simplified expressions for these inputs. The **k-Maps** for S & R are shown below.





So, we got S = D & R = D' after simplifying. The **circuit diagram** of D flip-flop is shown in the following figure.



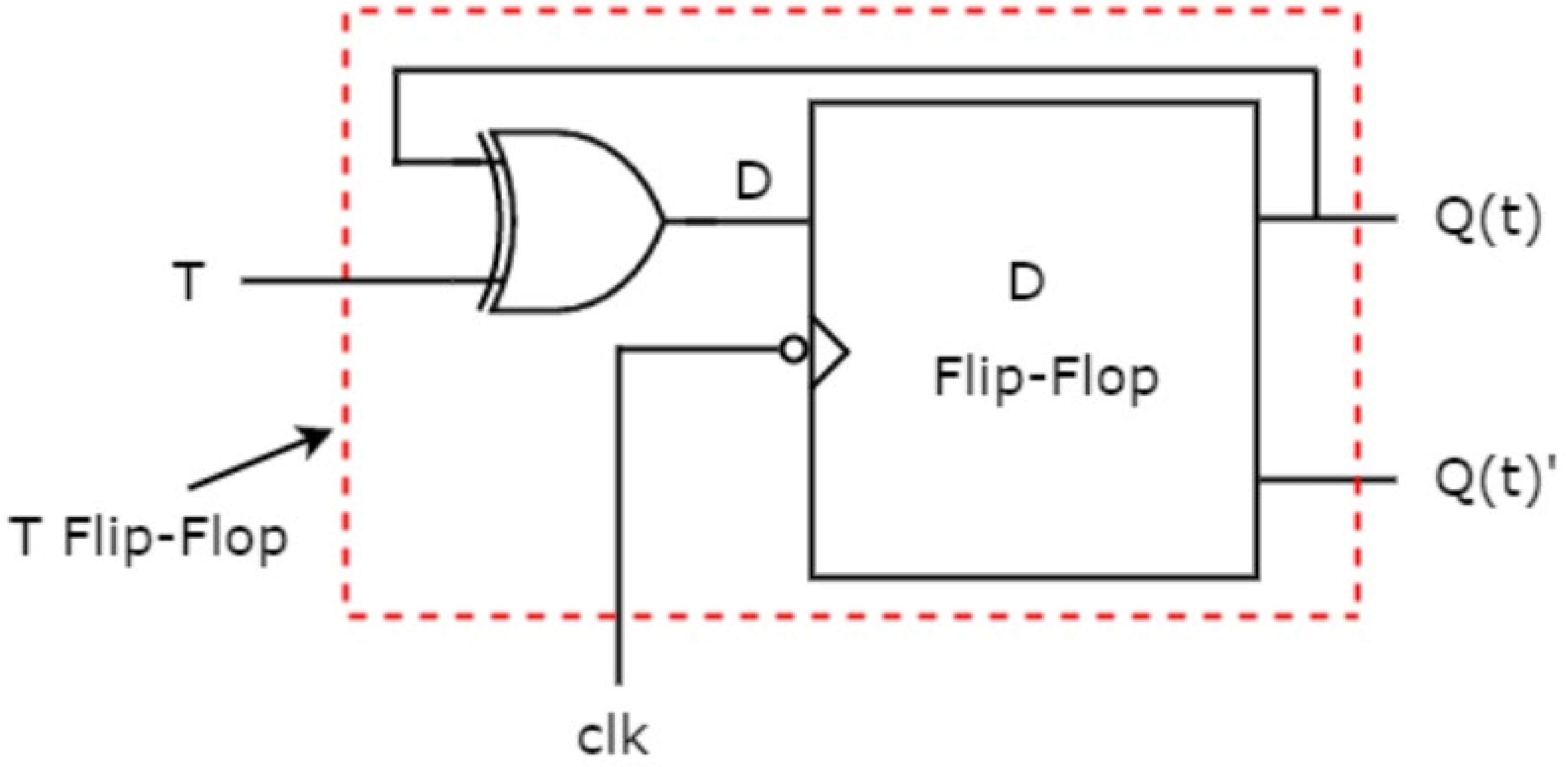
This circuit consists of SR flip-flop and an inverter. This inverter produces an output, which is complement of input, D. So, the overall circuit has single input, D and two outputs Qt & Qt’.

D flip-flop to T flip-flop conversion

Characteristic table of T flip-flop along with the **excitation input of D flip-flop**

T flip-flop input	Present State	Next State	D flip-flop input
T	**Q** t	**Q** $t+1$	**D**
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

$$D = T \oplus Q\,(t)$$


D
T
D
Flip-Flop
Q(t)
Q(t)'
T Flip-Flop
clk

Introduction Registers

- A FF is one bit memory cell. To increase the storage capacity we use multiple FF.
- A set of *n* flip-flops that can be used to store *n* bits of information, such as an *n*-bit number is called as a *register*.
- A common clock is used for each flip-flop in a register.
- We know that a given number can be multiplied by 2 if its bits are shifted one bit position to the left and a 0 is inserted as the new least-significant bit.
- Similarly, the number is divided by 2 if the bits are shifted one bit-position to the right.
- These can be achieved by shifting the contents of a register.

There are four main types of shift registers:

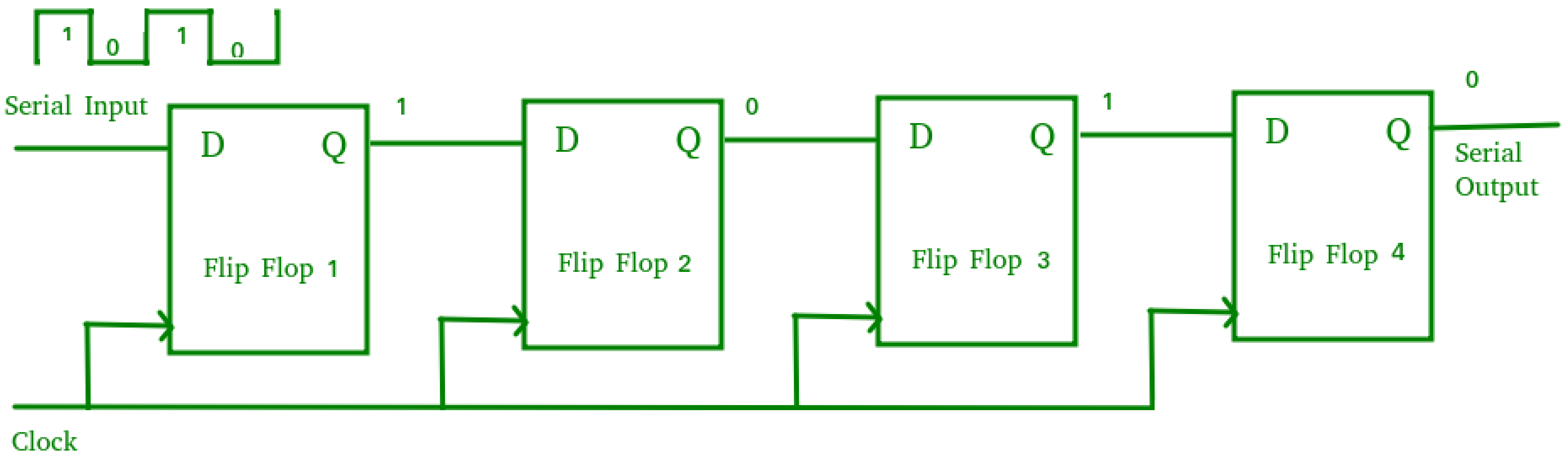
Serial-in serial-out (SISO): The data is shifted serially “IN” and “OUT” of the register, one bit at a time in either a left or right direction under clock control.

Serial-in parallel-out (SIPO): The register is loaded with serial data, one bit at a time, with the stored data being available at the output in parallel form.

Parallel-in serial-out (PISO): The parallel data is loaded into the register simultaneously and is shifted out of the register serially one bit at a time under clock control.

Parallel-in parallel-out (PIPO): The parallel data is loaded simultaneously into the register, and transferred together to their respective outputs by the same clock pulse.

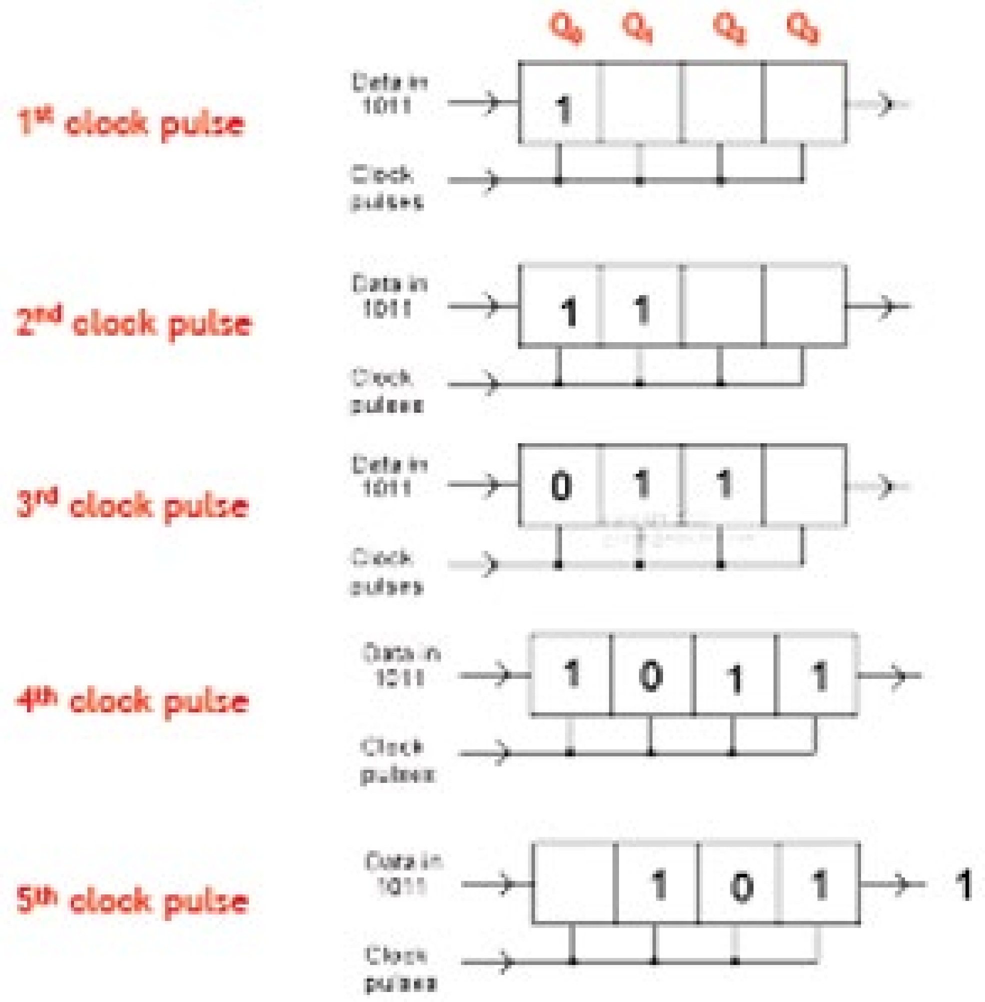
Serial In Serial Out (SISO) Shift Register



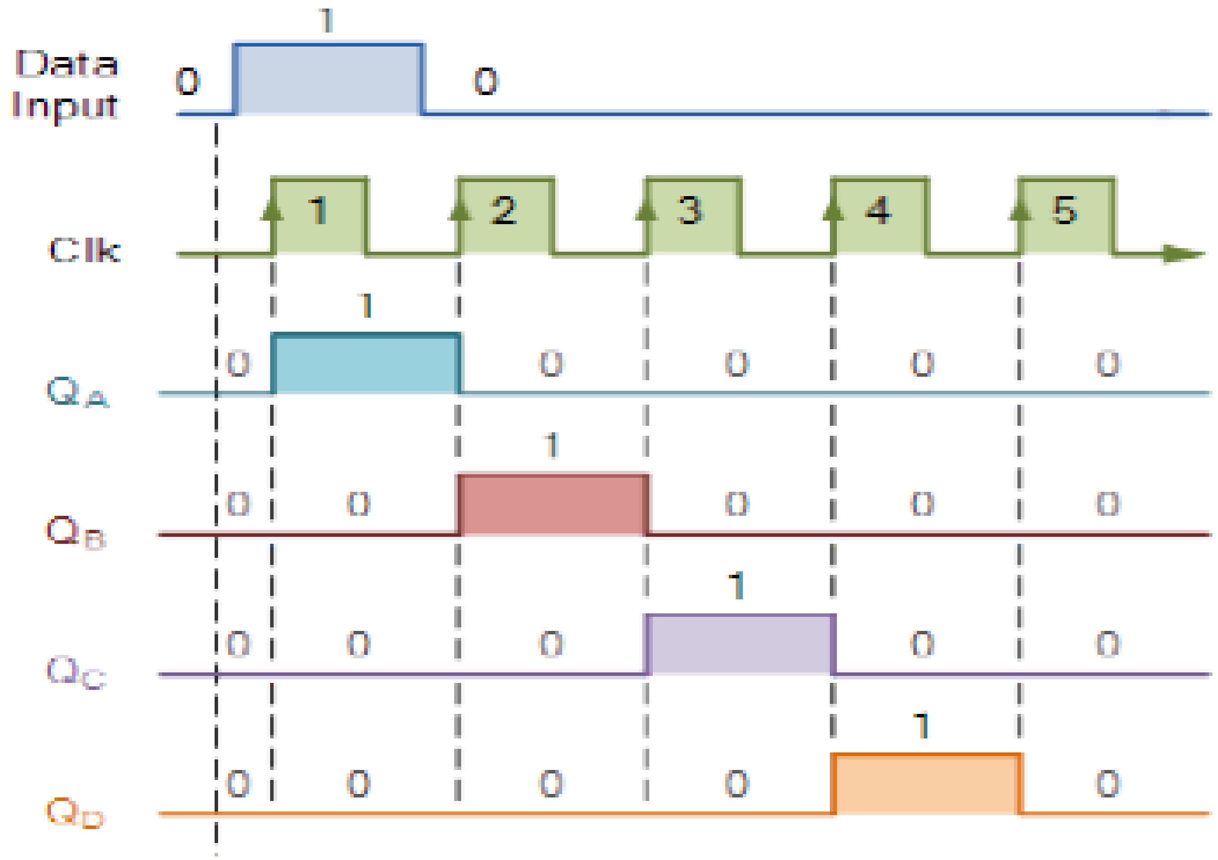
logical input 1011

Operation of the Shift-right Register

Timing pulse	Q_A	Q_B	Q_C	Q_D	*Serial output at* Q_D
Initial value	0	0	0	0	0
After 1st clock pulse	1	0	0	0	0
After 2nd clock pulse	1	1	0	0	0
After 3rd clock pulse	0	1	1	0	0
After 4th clock pulse	1	0	1	1	1

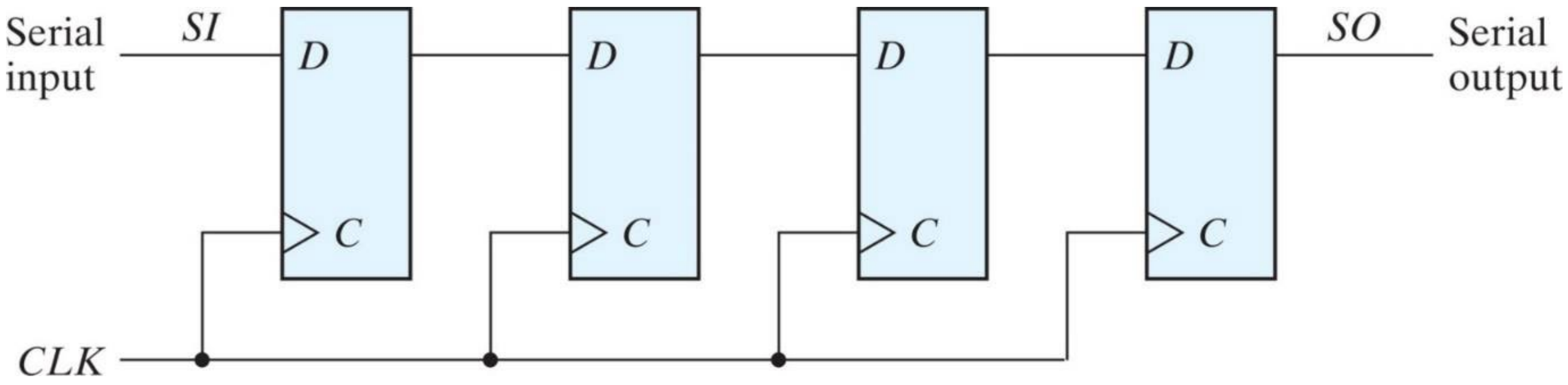


After 8th clock pulses, the register is CLEAR



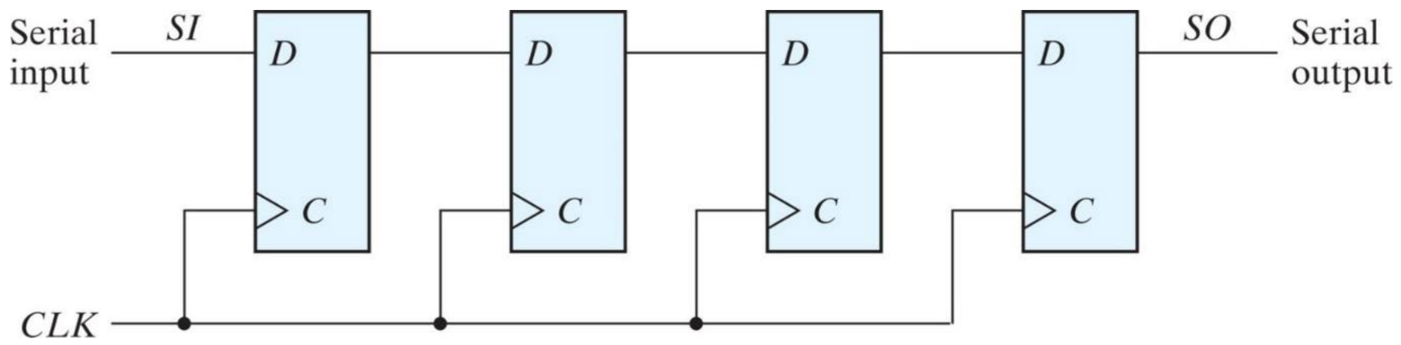
Verilog HDL – SISO (Structural Modeling)

```
module siso_str(clk,si,rst,so);
input clk,rst,si;
output so;
wire so;
wire qA,qB,qC;
dff_inst d1(clk,rst,si,qA);
dff_inst d2(clk,rst,qA,qB);
dff_inst d3(clk,rst,qB,qC);
dff_inst d4(clk,rst,qC,so);
endmodule
```

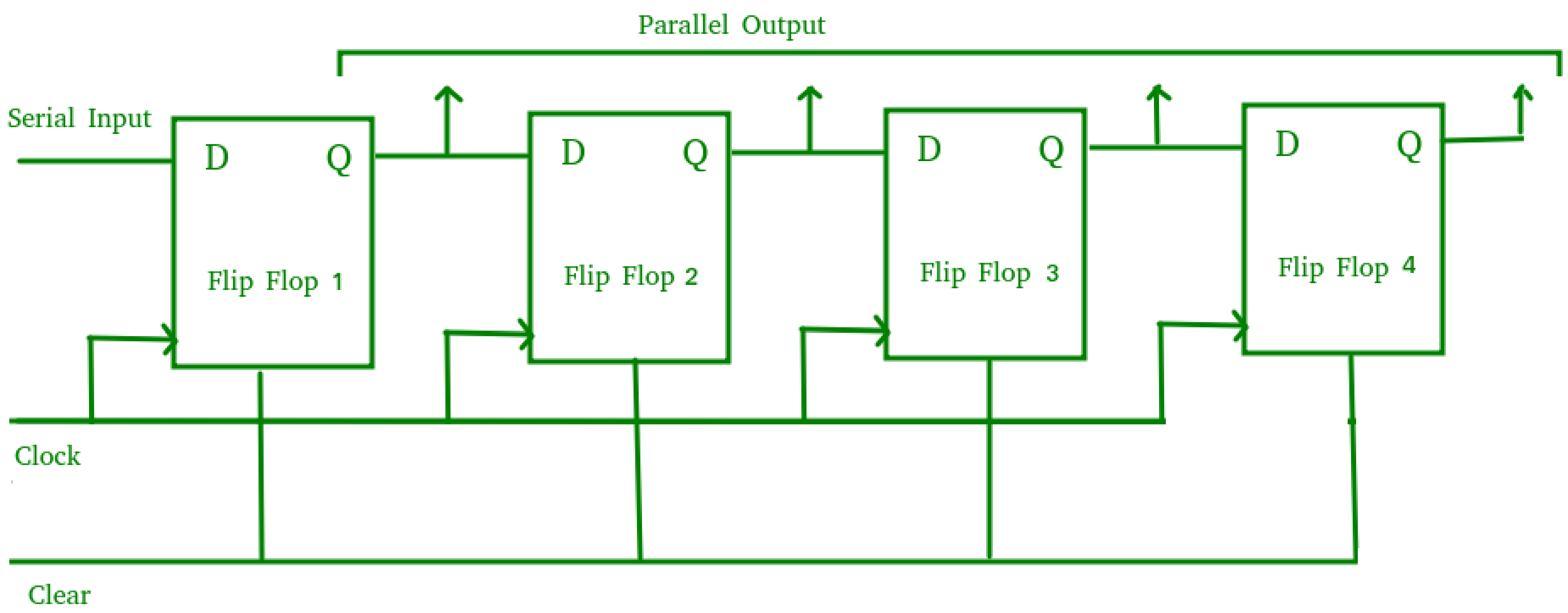


Verilog HDL – SISO (Behavioral Modeling)

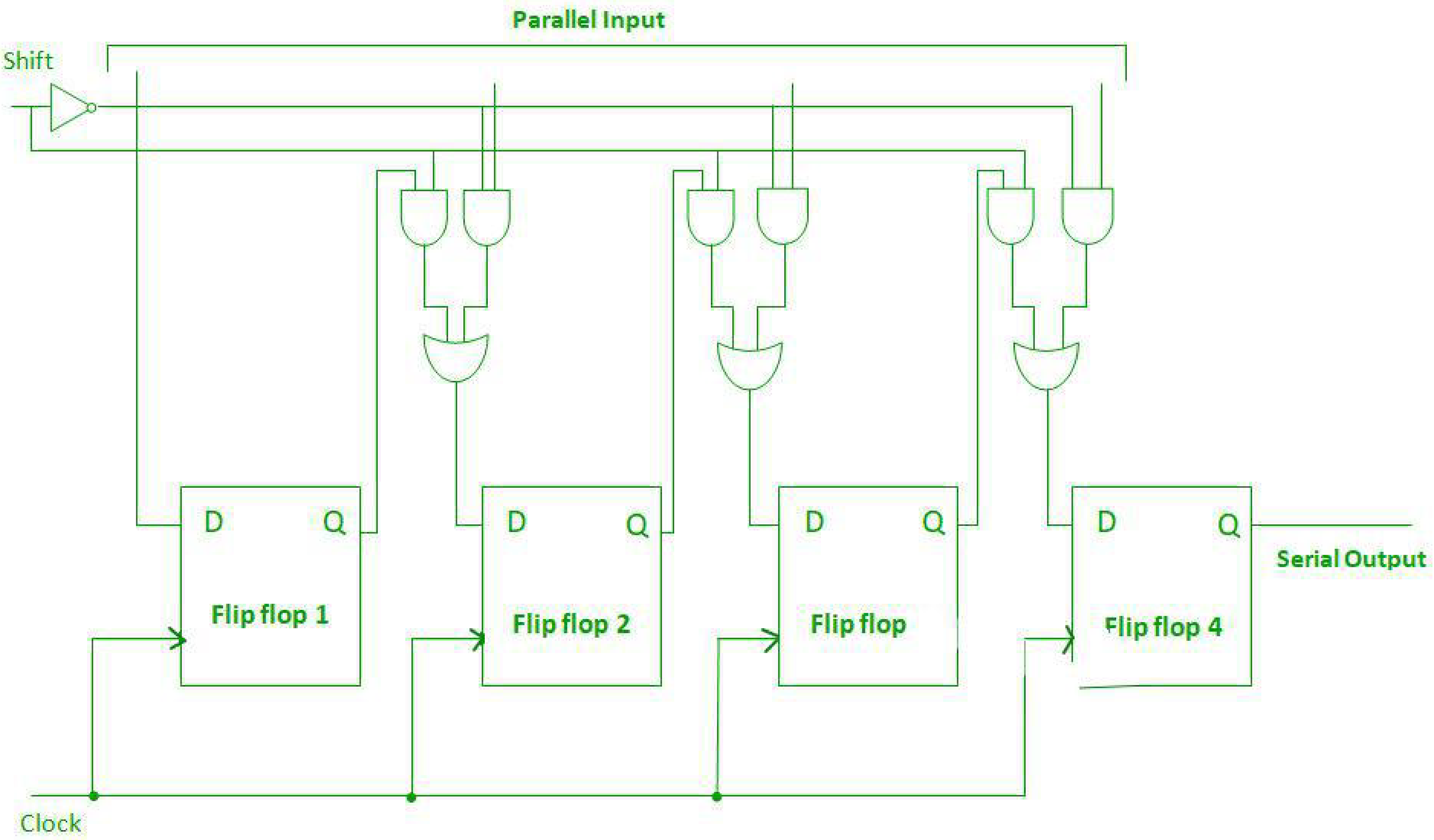
```
module siso_beh(clk,si,rst,so);
input clk,rst,si;
output so;
reg qa,qb,qc,so;
always @(posedge clk or posedge rst)
if(rst) begin
qa <= 1'b0; qb <= 1'b0; qc <=1'b0; so <= 1'b0;
end
else begin
qa <= si; qb <= qa; qc <= qb; so <= qc;
end
endmodule
```



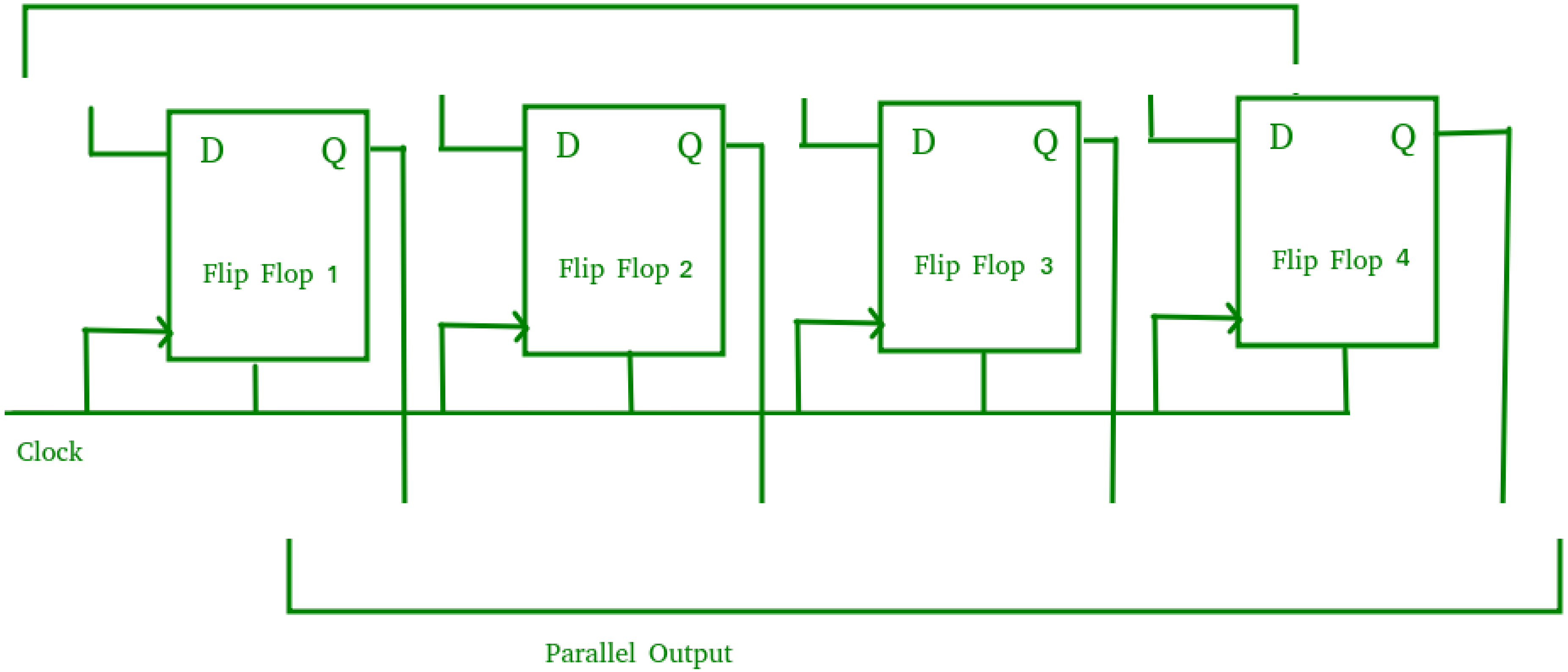
Serial In Parallel Out (SIPO) Shift Register



Parallel In Serial Out (PISO) Shift Register



Parallel In Parallel Out (PIPO) Shift Register

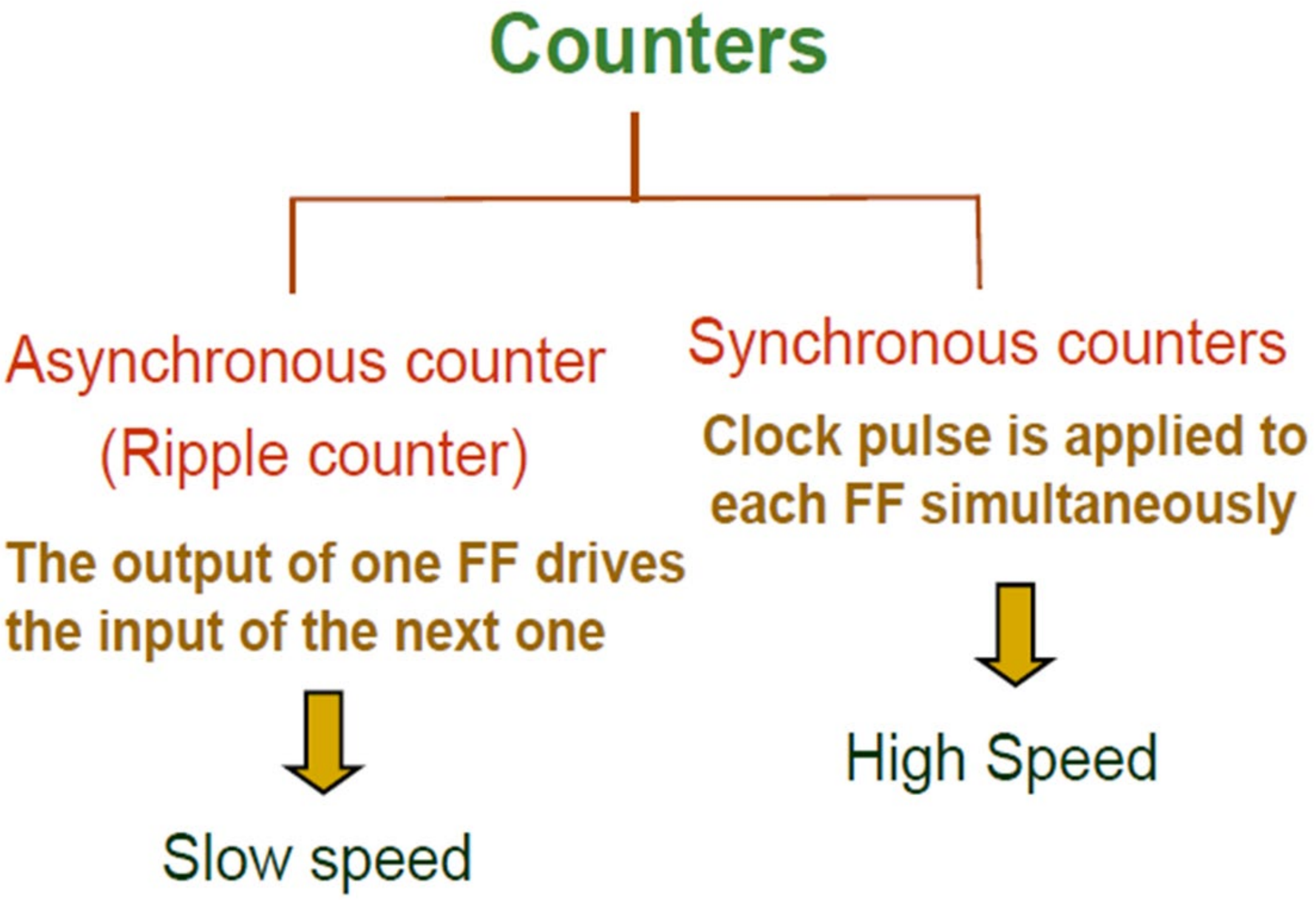


Counters

- A register that goes through a prescribed sequence of states upon the application of input pulses (clock pulse) is called a counter.

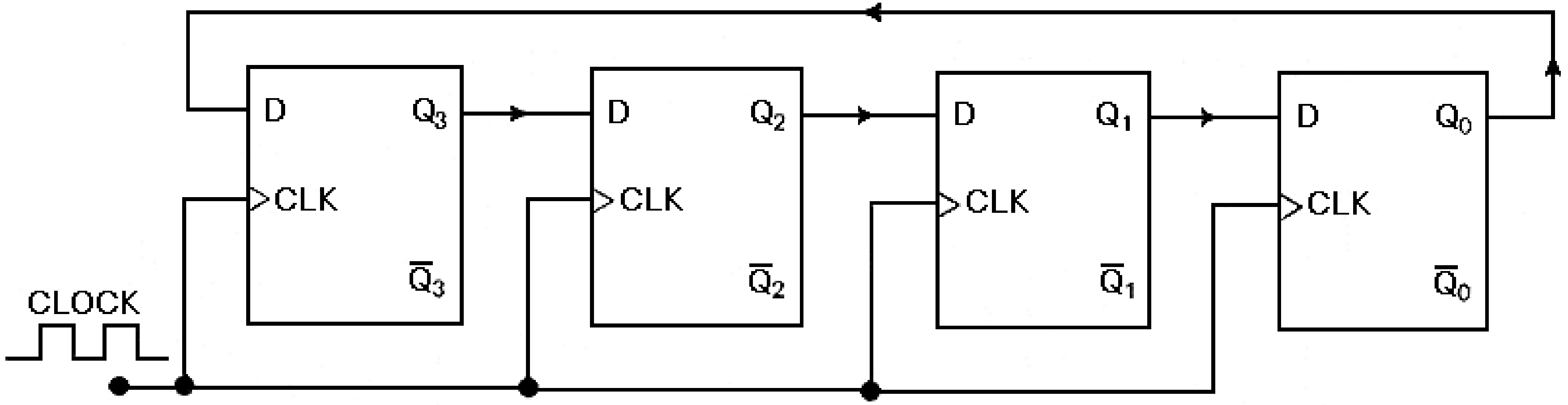
1. Ripple counters - Flip flop output serves as a source for triggering other flip flops.

2. Synchronous counters - All flip flops triggered by a single clock signal

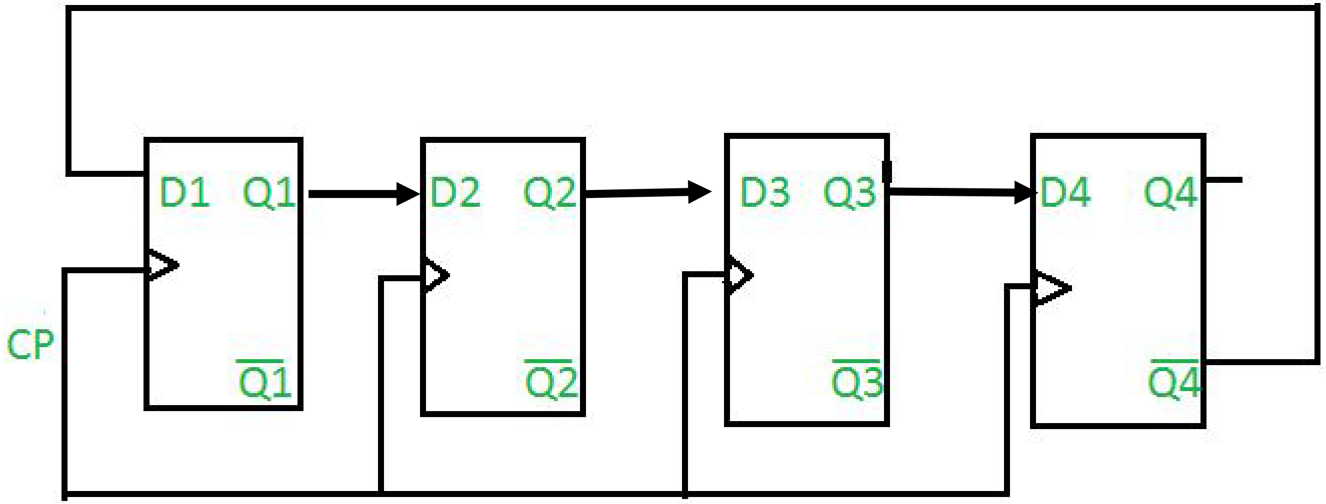


Ring Counter

- It is the special application of Shift register(circular shift register)
- The output of the last FF is given as the input of the first FF.



Johnson counters -also known as creeping counter, is an example of synchronous counter.



CP	Q1	Q2	Q3	Q4
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1
8	0	0	0	0